
MA214: APPLIED STATISTICS — LAB

Lab 0: Welcome to the labs

Module 1: Modeling and Linear Regression

- *Previously:* nothing — this is the first lab meeting.
- *This time:* the machinery of the labs, so that from Lab 1 onward none of it is new.
 - the two kinds of lab, and the semester schedule
 - what a lab must contain to count as *completed*
 - the tutorial system, end to end: run it, work it, submit it
- *Reading:* the syllabus, lab section.
- *Due today:* the *Tutorial 0* hash, on Blackboard before you leave.
- *Due before Lab 1:* the Lab 1 tutorial.

No statistics today.

Lab Objectives

🕒 By the end of today, you should be able to...

- say what happens in a lab, and which labs are which;
- state exactly what you must submit for a lab to count as complete;
- open a tutorial and run code inside it; and
- generate a submission hash and post it to Blackboard.

📋 Learning objectives

- **(Lab policy)** What a completed lab requires — attendance, tutorial hash, worksheet
- **(Lab tooling)** Running the tutorial container and submitting work

How the labs run

Who is in the room

- *Lab instructor:* Dr. Yongho Lim — ylim2@bu.edu
Office hours: Monday 1:00–2:00pm, Thursday 12:00–1:00pm,
CDS 408
- *Teaching fellows:* Sidharth Soundararajan (sidsound@bu.edu)
- *Learning assistants:* Cynthia Huang (chuang12@bu.edu)

Teaching fellow and learning assistant office hours: TBA, posted on the course page.

What the labs are for

Lecture is where the statistics is developed. Lab is where you *do* it: on real data, in R, with other people.

- You work in *groups*. Statistics is argued about, not recited.
- You use *R* and RStudio. The code is a means, not the point.
- You produce something every session — a worksheet, a deliverable.

★ Key idea

The lab grade is not about getting clean output. It is about whether you can interpret what the output says and say why it is reasonable.

Two kinds of lab

Skills labs (5)

Short, self-contained, hands-on.

- Prepare with an online tutorial
- Work an in-lab worksheet
- Graded completed / not completed

Project labs (2 projects)

Several weeks each, one group product.

- Project 1: video presentation
- Project 2: written report
- Graded E / S / R / N

Roughly half the semester is skills labs, half is project work. Both count toward your final grade, and they count in different ways — see the syllabus grading table.

The lab schedule

Week	Session	Focus
1	Lab 0	Welcome to the labs
2	Lab 1	Working with R and data
3	Lab 2	Regression models
4	P1-1	Project 1 launch and outline
5	P1-2	Project 1 data cleaning and EDA
6	P1-3	Project 1 modeling and diagnostics
7	P1-4	Project 1 presentation work session
8	P2-1	Project 2 launch: bootstrapping
9	P2-2	Project 2 simulation design
10	P2-3	Project 2 real data analysis
11	P2-4	Project 2 final work session
12	Lab 3	Bayes' rule
13	Lab 4	Conjugate families: beta-binomial
14	—	No lab
15	Lab 5	Bayesian linear regression

Note: Calendar dates follow your own section's meeting day — check your schedule and the course page.

What counts as a completed lab

The shape of a skills lab

Every one of the five runs the same way.

1. *Before lab.* Work the online *tutorial* for that lab. It practises the R on a different scenario, so that syntax is not what slows you down in the room.
2. *In lab.* Work the *worksheet* with your group, with the lab team circulating.
3. *Submit.* The tutorial's *hash* to Blackboard, and the worksheet as instructed for that lab.

★ Key idea

A skills lab counts as completed only when both arrive: the tutorial hash *and* the worksheet. One without the other is not.

What is required of you

- *Show up.* Lab attendance is graded. For an A you may miss at most one lab.
- *Come prepared.* Do the tutorial before the lab it belongs to, not after.
- *Submit both pieces* for each skills lab: hash and worksheet.
- *Carry your weight on projects.* Every deliverable is followed by a peer assessment form.

Caution — absences

An absence is excused only if you email the lab instructor in advance with your reason. Excused absences are not counted as missed. After the fact is too late.

The lab projects, in one slide

You will work in the same group on two projects.

- *Project 1* — a video presentation. Weeks 4–7.
- *Project 2* — a written report. Weeks 8–11.

Each is graded on the ESRN rubric: *E*xcellent, *S*atisfactory, *R*evisions needed, *N*ot assessable. An S earns full credit. An R or N must be revised and resubmitted once.

Both projects must reach S or E by the end of the course.

Note: More detail in the Week 4 lab, when Project 1 launches. Nothing to do about it today.

The tutorial system

Getting the tutorials running

The tutorials run in [Docker](#), so you install no R packages and set no working directory. Install Docker Desktop once, then for every tutorial:

Run the tutorial launcher that is available on the blackboard.

Then open `http://localhost:3838` in a browser (your launcher will open it for you). You will see a list of tutorials. Click the one you want to run.

- [Back to Main Tutorial Page](#) at the top of a tutorial returns to the picker.
- Nothing is downloaded and nothing is installed into R — the data lives inside the tutorial.

Inside a tutorial

- *Exercise box* — an editable code editor with a Run Code button. Run it as often as you like; you cannot break anything.
- *Blanks* — ___ marks something for you to replace.
- *Hint* — available from the start. Use it.
- *Solution* — released on a timer, normally after that tutorial's deadline. Do not wait for it.
- *Quiz* — multiple choice, with retries.



Caution — one box, one world

Each exercise box starts fresh. An object you create in one box does not exist in the next one.

Submitting a hash

At the bottom of every tutorial is a *Submit your work* section.

1. Check that you have attempted every exercise and quiz.
2. Type your *name* and *student ID*.
3. Click *Generate*.
4. Select the *whole* hash and copy it.
5. Paste it into that tutorial's submission form on *Blackboard*.
 - No hash appears? Answer at least one question first, and check that neither name box is blank.
 - Changed an answer afterwards? Generate a new hash and submit the newer one.
 - The hash encodes your answers and your ID. Do not post it anywhere public.

When you are stuck

In this order:

1. Read the error message, then read the question again.
2. Check spelling, capitalisation, brackets, quotation marks.
3. Compare your code with the worked example above it.
4. Open the hint.
5. Ask your group, then an LA, a TF, or the lab instructor.

Say what you already tried, and do not delete the broken attempt — it is the most useful thing you can show us.

★ Key idea

An error message is not a verdict on you. It is R telling you what it needs next.

Wrap-up

Key takeaways

- Five *skills labs* and two *projects*, roughly half the semester each.
- A skills lab is completed only when both the tutorial hash and the worksheet arrive — one without the other is not.
- The tutorial for a lab is done before that lab, not after.
- Attendance is graded, and an absence is excused only if you email *in advance*.
- Tutorials run in *Docker*. Download *Docker Desktop* and open it.
- *Today*: finish Tutorial 0 and submit its hash to Blackboard before you leave.
- Go over the R onboarding website for R and RStudio if you are new to R.
- <https://bu-intro-stats.github.io/MA213-214-R-onboarding-website/>